MIL-G-80159A
18 December 1974
SUPERSEDING
MIL-G-80159
9 July 1968

MILITARY SPECIFICATION

GENERATORS, ENDOTHERMIC ATMOSPHERE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers atmosphere generators used to change the chemistry of hydrocarbon gas to an endothermic type atmosphere for use in heat treating furnaces for metallurgy processes.

* 1.2 Classification. This specification covers atmosphere generators of the following types and sizes. The type and size supplied shall be as specified (see 6.2.1).

1.2.1 Types.

Type I - Gas Fired

Type II - Electrically heated

1.2.2 Sizes. The size shall be based on the cubic feet per hour (CFH) generator output of generated atmosphere, and should be selected from the sizes listed in Table I. If a size other than one listed in the table is required, the size in CFH shall be as specified.

TABLE I

Generator Output Capacity			
Size	CFH	Size	CFH
1	250	6	2000
2	500	7	2500
3	750	8	3000
4	1000	9	4000
5	1500		

FSC 3424

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2. APPLICABLE DOCUMENTS

* 2.1 Military publications. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION

MIL-F-3296	Forges, Furnaces, and Ovens (Exclusive of Space Heating and Cooking), Packaging of

STANDARDS

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of the specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments	Occupational Safety and Health Administration Standards

NATIONAL BUREAU OF STANDARDS (NBS) HANDBOOK H28

Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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JOINT INDUSTRIAL COUNCIL (JIC)

EGP - 1 Electrical Standards for General Purpose Machine Tools

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101.)

AMERICAN NATIONAL STANDARDS

ANS C96.1 Temperature Measurement Thermocouples

(Application for copies should be addressed to the American National Standards Institute, Dept. 969, 1430 Broadway, New York, NY 10018.)

FACTORY MUTUAL ENGINEERING CORPORATION (FM)

Handbook of Industrial Loss Prevention, Chapter 37

(Application for copies should be addressed to Factory Mutual Engineering Corporation, 1151 Boston-Providence Turnpike, Norwood, Mass. 02062.)

(Technical society and technical association standards and specifications are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

* 3.1 First Article approval. If First Article approval is required (see 6.2.1 and 6.3), inspection thereof shall be in accordance with Section 4. For production contracts, a First Article comprises a pre-production model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract. For single item procurements, the generator submitted to fulfill the terms of the contract or purchase order shall be considered the First Article.

* 3.2 Design. The generators shall be new and of the manufacturers current design. The generators shall be designed to produce a controlled atmosphere by cracking a mixture of hydrocarbon gas with air over a heated catalyst. The catalyst shall be housed in a retort surrounded by a refractory insulated heating chamber. The retort shall completely separate the heating elements of gases of the heating system from the catalyst. The atmosphere shall be produced by the reaction of the air-gas mixture ranging from 25 to 40 percent of perfect combustion at a temperature of at least 1750°F. The generator shall consist of a refractory insulated heating chamber, one or more catalyst filled retorts,gas-air pump with motor, carburetor mixing system, temperature control system,

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and all necessary gas proportioning regulators, pressure relief and control valves, safety shut-off valves, switches, gas and air flowmeters, primary and secondary pressure gauges, and the appropriate heating system for the type generator specified. All components shall be piped, wired and mounted on a common base and housing to form a package unit. Design of the generator shall provide for manually selecting any desired hydrocarbon gas-air ratio required to produce an atmosphere with an analysis specified herein. The mixing system shall maintain a constant air-gas ratio at the selected value, regardless of the rate of flow or variable atmosphere demand on the generator. Unless otherwise specified, the generator mixing system shall require manual starting, after the retort has heated to operating temperature. When required (see 6.2.1), the generator shall be provided with a control system to automatically actuate the mixing system, after retort has heated to operating temperature.

3.2.1 Type I. Unless otherwise specified, the Type I generators shall be designed to be heated by natural gas having a heat content of 1000 BTU per cubic foot and a supply pressure of 6-8 inches water column. When required, the generators shall be designed and equipped to operate on either a butane or propane fuel supply as specified (see 6.2.1).

3.2.2 Type II. The Type II generators shall be designed to be electrically heated, and shall operate from a single source primary power supply. Unless otherwise specified (see 6.2.1), the generators shall operate on 230 volt, 60 cycle, 3 phase current.

3.2.3 Safety devices. The generators shall be provided with an indicating excess temperature control pyrometer, equipped with fail safe thermocouple break protection. The circuits of the excess temperature control system shall be independent of the control circuitry. In the event the excess temperature control point is reached, the system shall function through an electrically operated valve and relay switches to shut off all gas supply, heat source, mixing pump motor, and combustion blower on gas fired systems. In addition to the excess temperature shut-off valve, a safety shut-off valve shall be provided with the necessary controls to shut off the gas supply in case of electrical, gas or combustion failure. A gas control cock shall be provided after the safety valve for manually checking the safety shut-off valve for leaks. Both a low pressure switch and a high pressure switch shall be provided between the safety shut-off valve and gas inlet to close the safety valve and cut off the gas supply in case the gas pressure varies more than plus or minus 10% of operating pressure. If a combustion blower is utilized on gas fired generators, a low pressure switch shall be installed in the combustion air line to shut down the generator in case of insufficient blower pressure. A normally closed motorized valve with a normally open auxiliary switch shall be installed in the gas line to the mixing system. The motorized valve shall open when the mixing pump is started and the auxiliary switch shall shut

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down the mixing pump in the event the motorized valve fails. The temperature control instrument shall be provided with an auxiliary temperature control circuit. This circuit shall be set at 1750°F, and shall be interlocked in the electrical circuit with the motor starter and motorized valve in the mixing pump to prevent starting the mixing pump or opening the gas supply to the retort until the desired cracking temperature is reached. If the temperature in the retort drops below the set point, the mixing pump shall stop, and the valve in the supply line shall close. An automatic fire check valve shall be provided between the retort inlet and mixing pump. In event of back fire, the fire check shall automatically shut off the gas-air mixture. An alarm bell and red warning light shall be provided to give both an audible and visual signal in the event the generator is shut down for any reason. If the generator is shut down by any safety devices, it shall remain down until restarted manually. The safety system of each generator shall conform to FM "Handbook of Industrial Loss Prevention", Chapter 37. Except as otherwise specified by the procuring activity, the generator shall comply with the standards promulgated under OSHA applicable to the generator manufacturer. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

3.2.4 Material. All material used in the fabrication of the generator and related equipment shall be new and of the quality necessary to produce a generator to meet the requirements described herein. Materials that are exposed to the internal environment of the generator chambers shall be compatible with the atmosphere and temperature conditions the materials will encounter.

* 3.2.5 Lubrication. Means shall be provided for lubrication of all sliding and rotating parts, except non-additive pack seal bearings. Lubrication systems shall be designed and built to prevent any contamination of atmosphere, components, material, or equipment. Lubrication points shall be readily accessible, and identified in the operators manual with type of lubricant and instructions for application.

3.2.6 Accessibility. All parts subject to wear, breakage, or distortion, and all parts which require periodic maintenance, shall be accessible for adjustment and replacement, as applicable.

* 3.2.7 Maintainability. Item configuration and adjustment features shall be designed to reduce maintenance operations to the simplest procedures compatible with a minimum of maintenance skills and common maintenance equipment. Instructions for disassembly, repair, adjustment, and reassembly will be concise and clearly defined in the operations and maintenance manuals.

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3.2.8 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or equipment.

3.3 Construction. The generators shall be constructed so that when installed and connected to the required supply sources, they can be used for any purpose described herein. All parts and accessory equipment shall be functionally suitable for the purpose intended. Repaired parts shall not be used.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. In no event shall such processes as welding, peening, plugging, or filling with solders or pastes be used on castings or forgings to reclaim any defective parts for use on the generators.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where those operations are required in the original design. These operations shall not be employed as repair measures for defective parts.

3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. Such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be rounded or beveled unless sharpness is required to perform a necessary function.

* 3.3.5 Painting. The generators shall be primed and painted with heat resistant paint in accordance with the manufacturer's standard commercial practice.

* 3.3.6 Threads. All threaded parts shall conform to NBS Handbook H28.

3.4 Electrical equipment. The generators shall be designed and furnished complete with all electrical motors, contactors, controls, relays, transformers, signal devices and switches pre-wired where possible for operation upon connection to source of power. All electrical equipment shall conform to JIC Electrical Standard EGP-1.

3.4.1 Motors. Motors shall have ball or roller bearings and all motors rated in excess of 1/2 horsepower shall be dual voltage, 230/460 volt, 3 phase, 60 cycle type, and unless otherwise specified (see 6.2.1), wired to operate on 230 volts.

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3.4.2 Electromagnetic interference control. When specified (see 6.2.1), the generators shall have electromagnetic interference control in accordance with the applicable requirements of MIL-STD-461.

3.5 Performance. The generators shall produce atmosphere at a constant discharge pressure of at least 6 ounces per square inch when operating in the range from 50 to 100 percent of rated CFH capacity. The generators shall provide for selecting any hydrocarbon gas and air ratio required to produce an atmosphere in equilibrium with any steel having a carbon content from 0.10 to 2.0 percent. The generators, operating at the richest setting shall produce an atmosphere with a composition by volume as follows:

Carbon Dioxide	0.0%
Oxygen	0.0%
Carbon Monoxide	20.0%
Hydrogen	40.0%
Methane	1.0%
Nitrogen	Balance
Dew Point Temperature	10° F

The generators, operating at the leanest setting, shall produce an atmosphere with a composition by volume as follows:

Carbon Dioxide	4.5%
Oxygen	0.0%
Carbon Monoxide	10.0%
Hydrogen	20.0%
Methane	0.0%
Nitrogen	Balance
Dew Point Temperature	60° F

3.6 Components and systems.

3.6.1 Shell and base. The generator shell shall be fabricated of at least 12 gauge sheet steel. The base shall be fabricated of angle or channel iron. The entire unit shall be reinforced with steel members

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and shapes welded to form a structurally sound unit that may be skidded or moved without damage. If legs are used, they shall be provided with bearing plates.

3.6.2 Retort and catalyst. The retort shall be either fabricated inconel not less than 3/16 inch thick, or centrifugally cast alloy containing at least 35 percent nickel and 15 percent chromium. The catalyst shall be nickel nitrate impregnated refractory capable of withstanding a temperature of at least 2300° F. The configuration of the catalyst shall prevent packing in the retort and shall provide sufficient surface area to produce the required reaction.

* 3.6.3 Insulation. The lining of the heating chamber shall consist of not less than 4½ inches of 2300°F insulating fire brick backed up with not less than 2½ inch thick magnesia block or preformed insulation. The quality and thickness of the lining shall be such that with the heating chamber temperature stabilized at 1850°F, the outside surface temperature of the chamber shall not exceed 100°F above ambient temperature. Ambient temperature is defined as 77°F ± 18°F in still air. Powdered, shredded, or granular material shall not be used for insulating purposes. When motar is used, it shall be compatible with the material being cemented together.

3.6.4 Heating systems.

3.6.4.1 Gas fired, Type I. The heating system for the Type I generator shall be complete, and capable of maintaining a combusion chamber temperature of at least 1950° F when operating at maximum capacity. Each pilot shall be provided with individual flame supervision. If a combustion blower is utilized in the design, the system shall be provided with an air filter, a manifold pressure gauge, and a pilot system complete with gas regulator and proportional mixer.

3.6.4.2 Electrically heated, Type II. Unless otherwise specified (see 6.2.1) heating elements for the Type II generator shall be of the ribbon or rod type with an alloy composition of 80 percent nickel and 20 percent chrome. Power density on the heating elements shall not exceed 15 watts per square inch of radiation surface.

3.6.5 Atmosphere gas-air mixing system. The atmosphere gas-air mixing system shall consist of a constant-pressure pump or blower, flow meters and carburetor system to maintain constant control ratio. The system shall be completely piped and assembled to proportion, mix, and pump the air-mixture through the retort. A separate gas flowmeter with micrometer needle valve to obtain precise dew point control shall be provided. The mixing system shall be provided with automatic by-pass regulation to maintain control

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ratios at all rates of flow within the capacity of the generator. An appliance regulator shall be provided in the gas supply line to the mixing system to modulate any gas pressure fluctuation. The intake air system shall be provided with an air filter.

* 3.6.6 Heat exchanger. Unless otherwise specified (see 6.2.1), each generator shall be provided with an atmosphere cooler or heat exchanger system. The system provided shall be capable of cooling the maximum generator output of atmosphere to a temperature not greater than 600° F in a time interval that will prevent any reverse reaction of the produced atmosphere. The produced atmosphere shall be further cooled to approximately 100° F above ambient temperature before entering the distribution lines or the bypass relief valve, when excess gas is produced.

3.6.7 Temperature controls. The temperature of the generator shall be automatically controlled by an electronic indicating controlling instrument equipped with thermocouple burn out protection. The instrument shall have an indicating calibrated scale length of not less than 6 inches and temperature range from 0° F through at least 50° F above the maximum operating temperature of the generator. The control indicator shall be provided with 2 pointers. One pointer shall be adjustable to the desired control temperature and one pointer shall indicate the actual temperature. This instrument shall be provided with an external control circuit to control the minimum safe gas cracking temperature specified in 3.2.3.

3.6.8 Thermocouples. Chromel-alumel type thermocouples shall be used as the temperature sensing devices. One thermocouple shall be used with the temperature controlling instrument, and one with the excess temperature control instrument. The thermocouples shall conform to ANS C96.1.

* 3.6.9 Atmosphere controlling, indicating, and recording instrument. When specified (see 6.2.1), the generator shall be provided with an instrument to automatically control, indicate, and record the atmosphere composition. The performance characteristics, the method of control (dewpoint or carbon content), and the minimum chart dimensions shall be as specified.

3.6.10 Instrument panel. Unless otherwise specified (see 6.2.1), the generator shall be provided with an integrally mounted instrument panel with all instruments, flow meters, gauges and controls mounted to conform to JIC EGP-1.

3.7 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

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3.8 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate, marked in accordance with MIL-STD-130 shall be securely attached to each generator. The nameplate shall contain the information listed below. If the generator is a special model, the model designation shall include the model of the basic standard generator and a suffix keyed to the manufacturer's permanent records.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Power input characteristics and ratings

Type of heating gas and total input requirement (CFH)

Output capacity (CFH)

Date of manufacture

Contract or order number

Federal Stock Number, National Stock Number, or Plant Equipment Code

U S

3.9 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.10 Workmanship. Workmanship of the generator and accessories shall be commensurable with the requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First Article inspection (see 4.3).

2. Quality conformance inspection (see 4.4).

4.3 First Article inspection. First Article inspection shall be performed on the first article sample submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the first article sample to pass the examination or one or more first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5 and the inspection in 4.7. Each item shall comply with all requirements.

4.5 Examination. The generator and equipment shall be examined for design, dimensions, construction, materials, components, electrical equipment, and workmanship, to determine compliance with the requirements of this specification.

4.6 Tests. All tests shall be conducted at an ambient temperature of 77° F ± 18° F.

4.6.1 Electromagnetic interference control test. Generators having electromagnetic interference control shall be tested in accordance with Table II of MIL-STD-461 to determine compliance with 3.4.2.

4.6.2 Operational test. The generator shall be prepared for and operated in accordance with the manufacturer's handbook of instructions until the refractory and insulation has cured and the combustion chamber has heated to atmosphere producing temperature. The mixing pump temperature control system shall be tested for conformance with 3.2.3 to prevent operation of the mixing pump when the combustion chamber temperature is below 1750° F. With the generator operating at 1800° F and producing atmosphere, the temperature control shall be adjusted to 1700° F. When the generator has cooled to a temperature below 1750° F, the mixing system shall stop and cut off the atmosphere flow. After this test, with the generator again operating at a temperature above 1800° F and producing atmosphere, the test in 4.6.2.1, 4.6.2.2 and 4.6.2.3 shall be performed.

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4.6.2.1 Thermocouple burnout. Each thermocouple lead shall be disconnected one at a time, to simulate thermocouple burnout. The temperature controller shall actuate controls to stop atmosphere flow, shut the generator down and sound the alarm.

4.6.2.2 Excess temperature control system. The excess temperature control system shall be tested by simulating an excess temperature condition to cause the system to operate. The system shall actuate controls to shut the generator down in accordance with 3.2.3. Accuracy of the excess temperature control system shall be within $\pm$ 25° F of control point.

4.6.2.3 Pressure switches and valves. The safety shut-off valve, low pressure and high pressure switch, low pressure switch in the combustion air line on gas fired types, and the auxiliary switch in the gas line to the mixing system shall be tested. Each safety device and system shall be tested by simulating a failure that normally would cause the device to operate. Each safety device shall perform as specified in 3.2.3.

4.6.3 Performance test. After the generator has been operated at 100 percent of rated capacity for at least 3 hours, an atmosphere composition analysis and flow rate analysis shall be performed. This test shall be performed with the generator operating at the richest setting, and again with the generator operating at the leanest setting. The flow rate shall be the cubic feet per hour for the size generator under test $\pm$ 5 percent. The atmosphere composition shall conform to the requirements specified in 3.5. The outside surface temperature shall be checked for compliance with 3.6.3. All meters and instruments used in the analysis shall be of known accuracy.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking for shipment and storage shall be inspected to determine conformance with the documents referenced in Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified in the contract or purchase order, preservation, packaging, and packing shall conform to Level A, B, or C requirements of MIL-F-3296 as specified (see 6.2.1), and marking shall be as specified therein.

6. NOTES

6.1 Intended use. Generators covered by this specification are intended to produce an endothermic type of atmosphere for use in controlled atmosphere furnaces, for surface protection of carbon steels, brazing, sintering, powdered metallurgy processes, and as a carrier gas for carburizing.

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6.2 Ordering data.

* 6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. First Article, when required (see 3.1).

d. Automatic starting, when required (see 3.2).

e. Heating chamber fuel supply, Type I only, if different (see 3.2.1).

f. Heating chamber power supply, Type II only, if different (see 3.2.2).

g. Specify exceptions and safety and health requirements (see 3.2.3).

h. Electric motor wiring, if different (see 3.4.1).

i. Electromagnetic interference control, if required (see 3.4.2).

j. Heating elements, if different (see 3.6.4.2).

k. Heat exchanger, if different (see 3.6.6).

l. Atmosphere controlling, indicating, and recording instrument, when required specify characteristics (see 3.6.9).

m. Instrument panel, if different (see 3.6.10).

n. Repair parts, when required (see 3.7).

o. Nameplate, if different (see 3.9).

p. First Article inspection, if different (see 4.3).

q. Quality conformance inspection, if different (see 4.4).

r. Level of preservation, packaging, and packing required (see 5.1).

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* 6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.3 First Article. When First Article approval is required under 3.1, the solicitation should adhere to the applicable portions of Section 1, Part 19 of the Armed Services Procurement Regulation.

* 6.4 Mercury prohibited. When the use of mercury or its compounds in the generator and its components and accessories is prohibited, procurement documents should include a statement to that effect.

* 6.5 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the Standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc. as applicable, specific requirements concerning guarding the point of operation, and other safety and health requirements should be specified.

* 6.6 Marginal notation. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document, based on the entire content, irrespective of the marginal notations and relation to the last previous issue.

Custodians:
Army - WC
Navy - SH
Air Force - 84

Review Activities:
Army - WC
Navy - SH
Air Force - 84
DSA - GS

User Activities:
Army - MU, WV
Navy - MC, AS

Preparing Activity:
DSA - IP

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED.

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